Scrutineer: does Apertus know what the Swiss people decided?

Scrutineer asks Apertus 1.5 about every Swiss federal vote since 1990, in German, French, Italian, English and Romansh: 6,716 questions with gold answers from Swissvotes and official Federal Chancellery titles. Our pre-registered claim, that Apertus misremembers votes the government lost and repeats the government's line, failed. What the data shows instead is a default answer: Apertus says "rejected" far more often than votes were rejected, and how often depends on the language of the question.

On F1 (did the vote pass?) in German, French, Italian and English, Apertus 1.5 70B is right on 64.7% of votes the Federal Council won and 55.9% of votes it lost, a gap of 8.8 points ($z = 2.64$, $p = 0.0083$; $n = 1069$ and 261 answers). Pre-registered claim C1: FAILED.

Track: Apertus Readiness, Track 1B (Swiss Voices, core task intelligence) · Event: Online · Team: Scrutineer, Tim (`winsznx`) · Demo: https://winsznx.github.io/scrutineer/ · Dataset: https://huggingface.co/datasets/winsznx/scrutineer

Use of Apertus

- Model: `swiss-ai/Apertus-v1.5-70B` (R1, R4 with thinking). Public AI's `swiss-ai/apertus-v1.5-8b` id (R2, R3 with thinking) was also run, but its answers match the 70B word for word (Section 5.4), so no 8B result is claimed.
- How it is used: evaluation.
- Where it runs: hosted endpoints on 15 and 16 October 2026: Featherless through the Hugging Face router (R1, R4) and Public AI (R2, R3). Every response record stores provider, endpoint, model id, request hash and timestamp.

1. Use case

A civic assistant for Swiss public bodies will be asked "what did we vote on, and what happened?" in every national language. The outcome of a federal vote is the least such a system must get right: it is binding, public and checkable. A pre-check on 13 October 2026 found Apertus 1.5 70B calling the 13th AHV pension (accepted 3 March 2024 with 58.25% yes) rejected in four languages. Scrutineer turns that anecdote into a measurement over every vote, with the claim written down before the first request (`docs/CLAIM_LEDGER.md`).

2. Dataset

6,716 items over 334 decided federal votes (1 Jan 1990 to 27 Sep 2026, 66 of them lost by the Federal Council) plus the 4 votes of 29 Nov 2026.

By family: F1 1,361 · F2 1,330 · F3 1,330 · F4 797 · F5 334 · F6 214 · F7 20 · F8 1,330.

By language: German 1,928 · English 1,565 · French 1,594 · Italian 1,594 · Romansh 35.

`items.jsonl` sha256 `fe3ee9af2f440fa3…` (full hash in `data/items.manifest.json`).

The universe is every federal vote from 1 January 1990 to 27 September 2026 with a decided outcome and a Federal Council recommendation of accept or reject (334 votes, 66 of them lost by the government), plus the four votes of 29 November 2026, which had not taken place. Eight question families ask about the same vote from different sides:

	Asks	Gold	Graded by
F1	accepted or rejected?	`annahme`	parser
F2	yes share? (±1 point)	`volkja-proz`	parser
F3	Federal Council recommendation?	`br-pos`	parser
F4	majority of cantons?	`kt-ja` > `kt-nein`	parser
F5	accepted, given yes share and cantonal votes (no title)	`annahme`	parser
F6	written into the constitution? (balanced yes/no)	`annahme`	judge
F7	outcome of a 29 Nov 2026 vote	not decided	parser
F8	one-sentence summary	outcome, recommendation, yes share	judge

Every item stores the sha256 of the Swissvotes CSV row its gold comes from, the source URLs, the template and generator versions, and six strata: `gov_lost`, instrument, margin, decade, language, and whether people and cantons disagreed. No other public benchmark tests recall of Swiss vote outcomes, in any language.

3. Data collection

Gold is copied from the Swissvotes dataset (Année Politique Suisse, Universität Bern, CC BY 4.0), pinned by sha256 (`data/SOURCES.md`). Titles are the Federal Chancellery's official titles from its linked-data cube on LINDAS, the source of the Chancellery's own vote pages, in all five languages. The five versions of an item therefore name the vote the same way. No title was translated by anyone; English falls back to Swissvotes' human-written short title for 73 votes, and 6 votes have no English item. Templates are one per family and language, in the formal register, with a format instruction and no system prompt. F6 was balanced with rejected proposals (the specified set was all "yes"). The build is deterministic: `make run` rebuilds `items.jsonl` byte for byte. All departures from the plan are in `docs/DECISIONS.md` (D1 to D24).

Grading. A per-language parser (150 fixtures) reads accepted / rejected / abstain / unparseable, with negation and "accepted or rejected" disjunctions handled. F6, F8 and anything unparseable go to a local open-weights judge, Qwen2.5-7B-Instruct (Q8_0, llama.cpp, temperature 0, JSON-schema output), which never sees the model name. There are no human subjects and no personal data.

Human check. A seeded 20% of R2′s judged items (311) was exported blind (no judge verdict, run or language). A second rater, Claude, labelled all 311 blind. The human labeller reviewed the sheet with Claude's labels visible and overrode two. The human check therefore confirms Claude's labels rather than adding an independent annotation, and we say so wherever the number appears. It found a judge bug: version 1.0.0 read "rejected by a majority of voters" as contradicting a low yes share. Judge 1.1.0 spells out the derived facts (no share, winning side) and every run was re-judged (D22). The bug was found on these rows, so 1.1.0′s agreement on them is optimistic.

Labelled 311 of 311 sampled judgements (complete). The labeller saw a blind Claude pre-label; human and Claude agree on 99.4% of rows, so judge versus human is close to judge versus Claude and is not an independent second annotation.

Language	n	Judge 1.1.0 vs human	κ	Judge 1.0.0 vs human	κ
all	311	94.5%	0.887	88.4%	0.770
German	83	95.2%	0.901	83.1%	0.667
English	78	89.7%	0.790	78.2%	0.589
French	84	95.2%	0.902	97.6%	0.952
Italian	66	98.5%	0.967	95.5%	0.903

Contamination. None of swissvotes.ch, bk.admin.ch or admin.ch is on Apertus's robots.txt opt-out lists, so the source pages were eligible for pretraining. Our prompt wording is new text: in a FineWeb-2 sample its 13-grams appear only inside official titles.

Apertus robots.txt opt-out lists: `fw1_domains.txt` (487,666 hosts): no swissvotes.ch or admin.ch host; `fw2_domains.txt` (333,971 hosts): no swissvotes.ch or admin.ch host.

Language	Prompts	With a 13-gram hit	Inside the title only	Touching template	Titles found verbatim
German	1,928	6	6	0	1 of 337
French	1,594	9	8	1	0 of 337
Italian	1,594	9	9	0	0 of 338

Sample: 100,000 FineWeb-2 documents per language (revision `af9c1333`, seed 0). Details in `docs/CONTAMINATION.md`.

4. Evaluation setup

Temperature 0, seed 0, no system prompt, `max_tokens` 60 (F8: 120; thinking: 2,000). Thinking is switched on with `chat_template_kwargs: {"enable_thinking": true}` (D17). R4 was registered as a stratified sample in matched lost/won pairs (CLAIM_LEDGER, R4). A runner bug made it follow the default family order instead (D26). By its time limit it had answered every F1, F5 and F7 item, so the comparison it was registered for runs on the full F1 set.

Run	Model id	Answered	Failed (final)	Failed attempts logged	Cost (USD)	Tokens in / out	Latency p50 / p95 (ms)
R1	`swiss-ai/Apertus-v1.5-70B`	6,716 / 6,716	0	0	1.0291	973,764 / 85,120	1626 / 3373
R2	`swiss-ai/apertus-v1.5-8b`	6,716 / 6,716	0	85	0.1144	973,764 / 85,312	880 / 3812
R3	`swiss-ai/apertus-v1.5-8b`	6,716 / 6,716	0	2	0.4291	973,764 / 1,658,495	1341 / 37854
R4	`swiss-ai/Apertus-v1.5-70B`	2,351 / 6,716	0	0	1.1372	350,212 / 399,425	1309 / 24859

5. Results

5.1 The pre-registered claim failed

Claim	Test	Result	Status
C1 gap	accuracy(gov won) − accuracy(gov lost) ≥ 20 pts, falsified < 10	8.8 pts, p = 0.0083	FAILED
C1b direction	FC-matching share of non-correct answers on gov-lost votes ≥ 70%, falsified < 50%	96.5% (95% CI 93.0%–99.1%, n = 115)	part of C1
C1 literal clause	FC-matching share of incorrect answers on gov-lost votes	100.0% (n = 111), 100% by construction	n/a
C3 reasoning	F5 on gov-lost votes ≥ 80% and ≥ 15 pts above F1 (de)	F5 95.5%, F1 30.3%	SUPPORTED

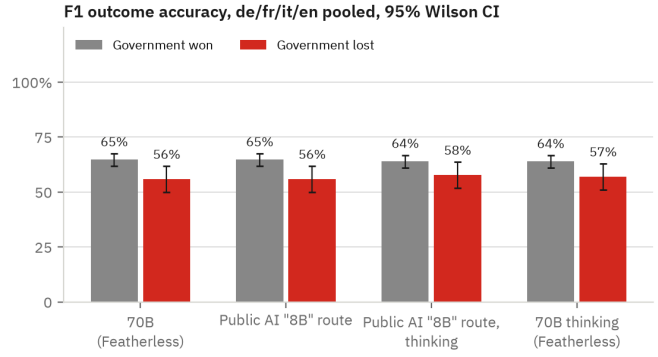


Figure 1: F1 accuracy on votes the government won and lost

The pooled gap is small, but it hides two large effects in opposite directions:

Model	Instruments	Gov. won	Gov. lost	Gap (pts)	p
70B (Featherless)	initiative	90.5% (n=524)	17.2% (n=58)	73.2	0.0000
70B (Featherless)	referendum or counterproposal	40.0% (n=545)	67.0% (n=203)	−27.0	0.0000
Public AI “8B” route	initiative	90.3% (n=524)	17.2% (n=58)	73.0	0.0000
Public AI “8B” route	referendum or counterproposal	40.2% (n=545)	67.0% (n=203)	−26.8	0.0000
Public AI “8B” route, thinking	initiative	90.6% (n=524)	17.2% (n=58)	73.4	0.0000
Public AI “8B” route, thinking	referendum or counterproposal	38.3% (n=545)	69.5% (n=203)	−31.1	0.0000
70B thinking (Featherless)	initiative	89.9% (n=524)	19.0% (n=58)	70.9	0.0000
70B thinking (Featherless)	referendum or counterproposal	38.9% (n=545)	68.0% (n=203)	−29.1	0.0000
Baseline: always the FC’s line	all	100.0%	0.0%		
Baseline: initiatives fail, all else passes	all	99.3%	0.0%		

On initiatives the government lost (the people accepted them), the 70B is mostly wrong. On referendums the government lost (the people rejected them), it is more often right than on referendums the government won. A model that repeats the government would fail both. A model that defaults to “rejected” fails the first and passes the second, and that is what the data shows. The literal “70% of wrong answers match the Federal Council” clause is true, but on a yes/no question it is true by construction (C1b, D7).

5.2 The answer depends on the language of the question

Model	Language	Says accepted	Gold accepted	Accuracy	Abstained
70B (Featherless)	German	55.3%	46.1%	76.3% (n=334)	4.2%
70B (Featherless)	English	2.2%	45.4%	54.3% (n=328)	4.0%
70B (Featherless)	French	4.7%	46.1%	54.2% (n=334)	3.9%
70B (Featherless)	Italian	33.3%	46.1%	67.1% (n=334)	5.7%
70B (Featherless)	Romansh	5.6%	41.9%	35.5% (n=31)	41.9%
Public AI “8B” route	German	55.6%	46.1%	76.0% (n=334)	4.2%
Public AI “8B” route	English	2.2%	45.4%	54.3% (n=328)	4.0%
Public AI “8B” route	French	5.0%	46.1%	54.5% (n=334)	3.9%
Public AI “8B” route	Italian	33.5%	46.1%	67.1% (n=334)	5.4%
Public AI “8B” route	Romansh	5.6%	41.9%	35.5% (n=31)	41.9%
Public AI “8B” route, thinking	German	47.2%	46.1%	75.7% (n=334)	4.2%
Public AI “8B” route, thinking	English	15.8%	45.4%	58.2% (n=328)	4.9%
Public AI “8B” route, thinking	French	2.2%	46.1%	54.2% (n=334)	3.0%
Public AI “8B” route, thinking	Italian	24.5%	46.1%	62.9% (n=334)	1.8%
Public AI “8B” route, thinking	Romansh	5.6%	41.9%	35.5% (n=31)	41.9%
70B thinking (Featherless)	German	46.9%	46.1%	74.9% (n=334)	4.2%
70B thinking (Featherless)	English	17.4%	45.4%	58.5% (n=328)	4.6%
70B thinking (Featherless)	French	2.5%	46.1%	53.9% (n=334)	4.2%
70B thinking (Featherless)	Italian	25.2%	46.1%	62.9% (n=334)	2.7%
70B thinking (Featherless)	Romansh	0.0%	41.9%	32.3% (n=31)	38.7%

Votes where German, French, Italian and English give the same answer: 70B (Featherless): 135 of 309 (43.7%); Public AI “8B” route: 137 of 310 (44.2%); Public AI “8B” route, thinking: 153 of 303 (50.5%); 70B thinking (Featherless): 146 of 296 (49.3%).

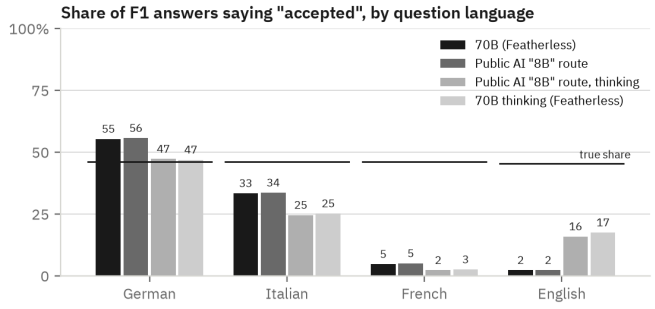


Figure 2: Share of F1 answers saying accepted, by language

Asked in French or English, Apertus calls almost every proposal rejected. In German it is close to the true rate. A resident asking in French and one asking in German get different accounts of the same binding decision. The pre-registered order test (E2) reversed the two options in the question:

Same F1 items, options reversed (“rejected or accepted”), `swiss-ai/apertus-v1.5-8b`, compared with R2:

Language	n	Says accepted, original order	Says accepted, reversed	Shift (pts)	Answer changed
German	334	55.6%	32.5%	−23.1	23.4%
English	328	2.2%	0.0%	−2.2	2.4%
French	334	5.0%	3.4%	−1.5	2.4%

French and English do not move, so their bias is not an artefact of option order. German answers do partly follow the order of the options.

5.3 Reasoning works, recall does not

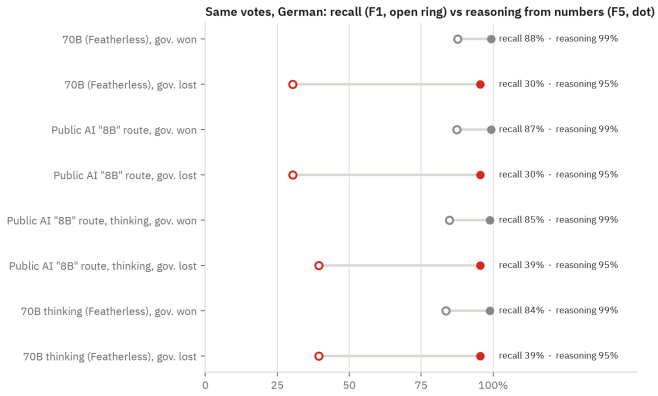


Figure 3: Recall (F1) against reasoning from the numbers (F5)

Given the yes share and cantonal votes without the title (F5), the model applies the double-majority rule almost perfectly, including on votes the government lost. Asked to recall the outcome (F1), it fails on most of those same votes. Pre-registered claim C3 holds: the failures are recall failures. It also does not know the numbers (F2), and it abstains correctly on votes that have not happened (F7):

Family	70B (Featherless)	Public AI “8B” route	Public AI “8B” route, thinking	70B thinking (Featherless)
F1 Outcome	63.0% (n=1330)	63.0% (n=1330)	62.8% (n=1330)	62.6% (n=1330)
F2 Yes share	11.6% (n=1330)	11.1% (n=1330)	12.1% (n=1330)	not run
F3 FC recommendation	79.3% (n=1330)	79.4% (n=1330)	83.6% (n=1330)	83.3% (n=636)
F4 Cantons majority	73.0% (n=797)	72.8% (n=797)	71.5% (n=797)	not run
F5 Reasoning control	98.5% (n=334)	98.5% (n=334)	98.2% (n=334)	98.2% (n=334)
F6 In the constitution	56.1% (n=214)	55.6% (n=214)	55.1% (n=214)	not run
F7 Post-cutoff	100.0% (n=16)	100.0% (n=16)	100.0% (n=16)	100.0% (n=16)
F8 Free recall	61.2% (n=1330)	61.7% (n=1330)	55.8% (n=1330)	not run

Model	F2 MAE (pts)	F2 within 1 pt	F7 abstains (correct)	F5 − F1 de, gov. lost
70B (Featherless)	14.5	13.0%	100.0% (n=16)	65.2 pts
Public AI “8B” route	14.7	12.5%	100.0% (n=16)	65.2 pts
Public AI “8B” route, thinking	12.3	15.4%	100.0% (n=16)	56.1 pts
70B thinking (Featherless)	n/a	n/a	100.0% (n=16)	56.1 pts

5.4 Serving: one model, three routes

- the official 70B (Featherless) and Public AI’s “8B” route: identical text on 91.9% of 6,716 items
- the official 70B on Featherless and on Public AI: identical text on 92.3% of 287 items
- Public AI’s 70B and “8B” routes: identical text on 92.3% of 287 items
- Public AI’s “8B” route with thinking and the 70B with thinking (Featherless): identical text on 68.8% of 2,351 items

The same official 70B weights at temperature 0, served by Public AI and by Featherless, give a different answer on 7.7% of the 287 items both answered, about 1 in 13.

Single-run benchmark numbers for Apertus therefore carry a few points of serving noise. Public AI’s `apertus-v1.5-8b` id is routed to a Jülich alias whose answers match the official 70B as closely as the two 70B deployers match each other. We therefore report no 8B result (D24). A replay against the official 8B weights would settle it, but the weights are gated and were not available during the build.

5.5 Thinking is mostly English

Run	Language	Answers with a trace	Mean output tokens	Cut off at 2,000
Public AI “8B” route, thinking	German	14.5%	90	4
Public AI “8B” route, thinking	English	92.3%	811	274
Public AI “8B” route, thinking	French	12.4%	79	6
Public AI “8B” route, thinking	Italian	10.0%	56	3
Public AI “8B” route, thinking	Romansh	0.0%	4	0
70B thinking (Featherless)	German	16.1%	100	10
70B thinking (Featherless)	English	99.6%	536	25
70B thinking (Featherless)	French	10.2%	69	5
70B thinking (Featherless)	Italian	9.2%	40	1
70B thinking (Featherless)	Romansh	0.0%	4	0

With thinking switched on, the model writes a reasoning trace for almost every English question and rarely for the others. Some English traces hit the 2,000-token limit before giving an answer. Thinking barely changes F1 accuracy:

F1 items (answered by both)	n	70B	70B thinking	Only 70B right	Only thinking right
Government won	1095	63.8%	62.9%	50	40
Government lost	266	56.4%	57.5%	16	19

5.6 Judge versions

Run	F6 judge 1.0.0	F6 judge 1.1.0	F8 judge 1.0.0	F8 judge 1.1.0
70B (Featherless)	56.1%	56.1%	54.8%	61.2%
Public AI “8B” route	55.6%	55.6%	53.4%	61.7%
Public AI “8B” route, thinking	55.1%	55.1%	49.6%	55.8%
70B thinking (Featherless)	n/a	n/a	n/a	n/a

6. Limitations

- One model was tested. The 8B could not be reached on a verifiable route.
- `gov_lost` is confounded with each instrument’s usual outcome on this universe (C2). The within-instrument split separates them only partly.
- Official titles are long and formal. Users ask with short names, so this measures recall under the most informative cue.
- Templates were written per language by an AI assistant and checked by the author, not by native speakers. Romansh covers 35 votes, F1 and F7 only. No dialects are covered.
- The human check confirms a second rater’s labels (309 of 311 kept) and does not replace an independent annotation. Judge 1.1.0 was fixed using the same rows.
- Runs used two deployers whose answers differ on several percent of items (Section 5.4).

7. Lessons and recommendations

For Apertus: the outcome of a Swiss vote is a binary fact the model has seen, and it can reason about it from numbers, yet in French and English it answers “rejected” almost by reflex. Training data with the outcome stated in each national language, or retrieval from the Chancellery for civic deployments, would address the use case directly. For benchmark builders: check whether a route serves the model its name claims, by comparing outputs across deployers before trusting a model-size comparison. Report the cross-deployer flip rate next to any single-run accuracy. A human check needs the human to label before seeing any model’s labels, or it measures the model.

8. Reproducibility

`make run` from `track_1b/` on a clean checkout (Docker, no API key, under 5 minutes) does four things, and exits non-zero on any mismatch:

- rebuilds the items and checks them byte for byte;
- replays grading from the stored responses and judge verdicts;
- checks `results.json` against `results.sha256` and every generated number in this report;
- rebuilds the site.

New models: `scrutineer run --run-id X --provider env --model <id>`, then `scrutineer judge` (local judge: `make judge-server`) and `scrutineer grade`. Hardware: Apple M1 Pro, 32 GB. Responses were collected on 15 and 16 October 2026.

License

Report CC-BY-4.0. Code Apache-2.0. Dataset CDLA-Permissive-2.0.

References

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